Chapter I

INTRODUCTION

In today's digitally-driven environment, organizations that handle financial transactions and member services must adopt efficient information systems to ensure accuracy, transparency, and operational effectiveness. Without integrated digital solutions, processes such as data entry and recordkeeping often remain manual and prone to errors, resulting in inefficiencies, delays, and inconsistencies in organizational performance (Sunankalijaga, 2024). These challenges are especially critical in cooperatives where financial and product transactions occur daily and require precise documentation.

Information management systems that are web-based have been shown to improve record accuracy and streamline workflows by replacing paper-based procedures with automated processes (Tanaman et al., 2023). By consolidating key functions into a unified platform, these systems support real-time data access, reduce duplication of records, and enhance reporting capabilities. For cooperatives like Bislig Chambers Multi-Purpose Cooperative (BICHAMCO), which currently relies on manual encoding and documentation for loan applications and inventory tracking, adopting an integrated digital solution can significantly improve service delivery and reduce processing time.

Centralized digital platforms also enhance monitoring and decision-making by providing managers with up-to-date information and reducing the time spent on retrieving manual records (IJRSI, 2025). Additionally, research on cooperative technology adoption indicates that digital systems enable

organizations to better engage members and strengthen institutional transparency, particularly when members are permitted access to their own financial information online (Habamenshi & Ernest, 2024).

1.1 Background of the Study

Bislig Chambers Multi-Purpose Cooperative (BICHAMCO) is an accredited multi-purpose cooperative located in Bislig City. It provides loan services to its members and operates grocery and rice trading activities to support their financial and daily needs. As a cooperative that handles financial transactions and product sales, accurate record management and efficient monitoring systems are essential to ensure transparency, accountability, and smooth daily operations.

BICHAMCO currently uses a system to manage loan records; however, the loan application process remains partially manual. Members are required to fill out paper-based forms, and staff manually encode the information into the system. This process increases processing time and creates opportunities for encoding errors and misplaced documents. Studies have shown that manual data entry and paper-based documentation often result in inefficiencies, delays, and data inconsistencies that affect organizational performance (Sunankalijaga, 2024). When records are not directly encoded by users, the risk of duplication and inaccurate information increases.

Beyond loan services, the cooperative's grocery and rice trading operations rely entirely on manual processes. Sales transactions, inventory monitoring, and member purchases are documented on paper. Manual inventory systems often lead to inaccuracies, and difficulties in tracking stock levels, particularly in environments with high daily transaction volumes (Tanaman e al., 2023). The absence

of automated inventory monitoring slow down the maintenance of real-time stock updates and the generation of accurate sales reports.

Considering that both loan services and trading operations involve member transactions and financial records, the absence of an integrated platform makes it difficult for management to monitor activities comprehensively. Delays in report generation, errors in data recording, and difficulty in accessing updated information can affect decision-making and member trust.

In response to these challenges, this study proposes the development of a Web-Based Cooperative Loan and Information Management System for BICHAMCO. The system is designed to digitize loan applications, computerize grocery and rice inventory management, and consolidate all records into a single platform.

In addition, the inclusion of a Member Portal will enable members to apply for loans online, view their loan balances, monitor transaction history, and make payments through the system. New members may complete a pre-application online; however, they are still required to visit the cooperative office for document verification and mandatory training before full access is granted. By integrating administrative functions and member access within one system, the proposed solution seeks to improve operational efficiency, enhance data accuracy, and strengthen monitoring and decision-making processes within the cooperative.

1.2 Statement of the Problem

This study investigates the operational challenges of Bislig Chambers Multi-Purpose Cooperative (BICHAMCO), emphasizing inefficiencies and limitations in current processes. Specifically, it aims to answer the following questions:

1. How can the cooperative address the delays and errors caused by manual loan application processing?
2. How can inaccuracies and difficulties in monitoring grocery and rice trading transactions be minimized?
3. How can members gain accessible, real-time information about loans and transactions in the absence of an online portal?
4. How can the cooperative evaluate the quality and performance of its operations and system?

1.3 Objectives of the Study

General Objective

The study aims to develop a Web-Based Cooperative Loan and Information Management System that addresses operational inefficiencies, ensures accurate record-keeping, and evaluates software quality based on ISO/IEC 25010 standards in BICHAMCO's loan and trading operations.

Specific Objectives

Specifically, the study seeks to:

1. Implement a digital loan application and payment monitoring module to reduce delays and errors in manual loan processing.
2. Develop a computerized grocery and rice trading module to minimize inventory inaccuracies and improve daily sales tracking.
3. Establish a Member Portal that provides members with real-time access to loan and transaction information.
4. Integrate a system evaluation framework using ISO/IEC 25010 to assess software quality characteristics, including functionality, reliability, usability, efficiency, maintainability, and security.

1.4 Significance of the Study

The significance of this study lies in its potential to improve the operations, clarity, and reliability of records at Bislig Chambers Multi-Purpose Cooperative (BICHAMCO).

Cooperative Staff and Management: The system will simplify the loan application process, reduce manual paperwork, and make it easier to track member information, loan payments, and grocery and rice trading records. This will help staff manage daily operations with fewer delays and errors.

Board of Directors and Committees: Committees such as the Credit and Investigation Committee and the Audit Committee will have faster access to accurate information, allowing them to monitor loans and member transactions more effectively and make informed decisions.

Cooperative Members: Through the Member Portal, members can apply for loans online, view their loan balances and payment history, monitor their purchases, and make payments. This provides easier access to records and reduces the need for frequent office visits.

Future Researchers and Developers: This study serves as a reference for future researchers and system developers interested in designing cooperative management systems or improving operational processes, offering insights into combining loan and trading management in one platform.

1.5 Scope and Limitation of the Study

Scope

The system is designed to improve the operations of BICHAMCO by providing an integrated platform for managing loans, member transactions, and cooperative business activities. The study focuses on the design and development of an Integrated Loan and Cooperative Management System with the following features:

1. Loan Management

- Pre-registration of new members (walk-in and online pre-application)
- Digital loan application and evaluation for existing members
- Loan approval and processing
- Tracking of loan payments and automatic calculation of penalties
- Maintenance of loan history for each member
- Generation of loan reports for monitoring and auditing

2. Grocery and Rice Trading

- Management of grocery and rice inventory, including stock levels and updates.
- Recording of daily sales transactions for all products.
- Recording of member purchases, whether paid in cash or on credit.
- Linking purchases to member profiles to calculate year-end benefits and incentives.
- Generation of comprehensive sales and inventory reports.

3. Member Portal

- Online loan application for existing members
- Viewing of loan details, balances, due dates, and payment history
- Online loan payment feature using GCash QR Code for proof of payment only.
- Viewing of grocery and rice purchase history
- Monitoring of transactions contributing to year-end benefits
- Pre-application for new members, subject to in-office verification and training.
- Uploading of required scanned documents for pre-approval.

4. Administrative Functions

- Management of staff and administrators and staff
- Role-based access control for system features
- Management of scanned member documents
- Data backup and basic security management
- Generation of consolidated reports for loans and trading

5. General Features

- Viewing of loan details, balances, due dates, and payment history
- Easy search and retrieval of member and transaction records
- Designed for on-site use within the cooperative via desktop or tablet devices

Limitation

The study is subject to the following limitations:

- New members must visit the cooperative office to submit required documents and complete training before full approval and portal access.
- Physical document submission is required; the system stores only scanned copies uploaded by members for pre-approval of loan officer.
- The system does not include a dedicated mobile application and is accessible only through web browsers.
- The system focuses only on Loan Management and Grocery/Rice Trading; other cooperative operations are not included.

- Advanced accounting features such as payroll, supplier management, and full financial statements are not included.
- Online loan payments are accepted exclusively through GCash via QR code, as the cooperative's only approved digital payment method.

1.6 Definition of Terms

- **Loan** – A financial product provided by Bislig Chambers Multi-Purpose Cooperative (BICHAMCO) to its members, allowing them to borrow a specific amount of money with an agreed schedule for repayment, including applicable interest.
- **Penalty** – A fee automatically imposed by BICHAMCO when a member fails to make a loan payment by the designated due date, calculated according to the cooperative's policies.
- **Pre-Application** – The initial online submission completed by a prospective member of BICHAMCO, which allows the cooperative to review basic member information. Full membership and system access are granted only after the in-person submission of documents and completion of mandatory training.
- **BICHAMCO** – Bislig Chambers Multi-Purpose Cooperative located in Bislig City that serves as the subject of this study, offering loan services and trading operations.
- **Cooperative** - An organization owned and operated by a group of individuals for their mutual benefit, providing financial and commercial services to its members.
- **Web-Based System** - A software application that is accessed through a web browser over a network, allowing users to perform tasks online without installing software locally.

- **Member Portal** - An online interface that allows cooperative members to access their personal records, apply for loans, and monitor transactions.
- **GCash QR Code Payment** - A digital payment method where members scan a QR code using GCash to complete loan payments, serving as the cooperative's exclusive online payment method.
- **Automation** - The use of technology to perform tasks with minimal human intervention, reducing manual work and errors.
- **Digital Loan Application** - An online process where members submit loan requests electronically instead of using paper-based forms.

Conceptual Framework

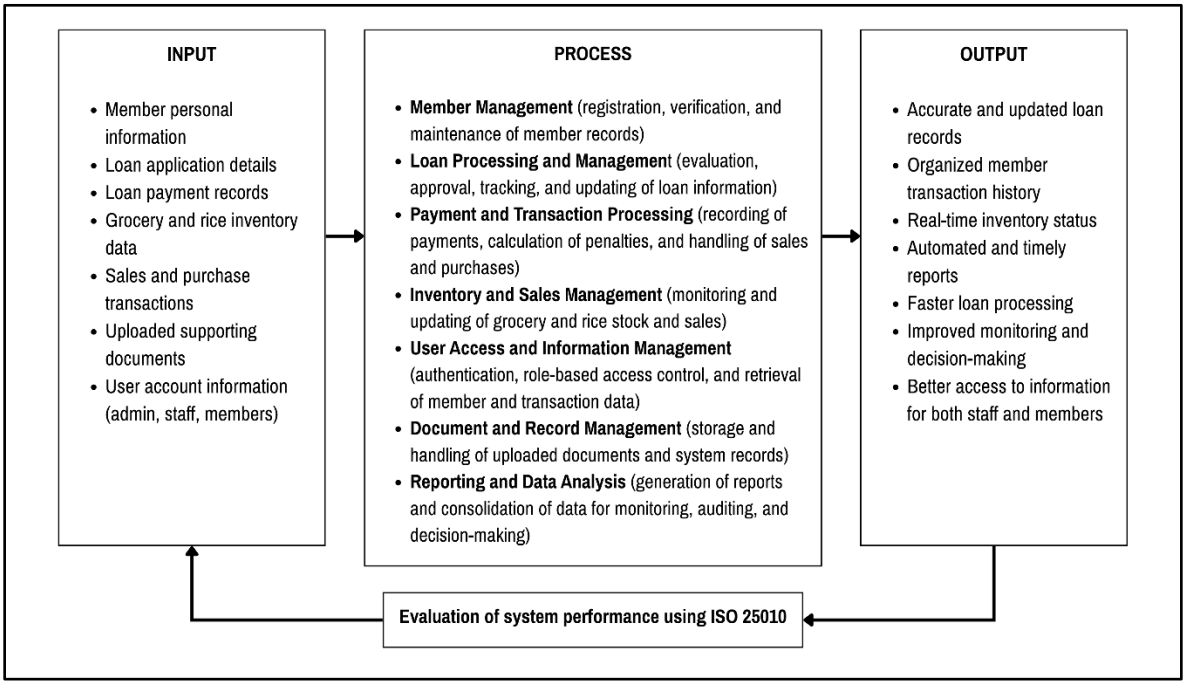

Figure 1. Conceptual Framework of the Study (IPO Model)

The conceptual framework of the study is illustrated in Figure 1, following the Input-Process-Output (IPO) model. It shows how inputs such as member information, loan applications, and grocery/rice transaction data are processed through the proposed web-based system to produce outputs, including digital records, reports, and accessible member services.

Chapter II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter reviews relevant literature and studies on web-based information management systems, cooperative management practices, and digital transformation in multi-purpose cooperatives to establish the theoretical and empirical foundation for the design and development of a Web-Based Cooperative Loan and Information Management System for Bislig Chambers Multi-Purpose Cooperative (BICHAMCO).

2.1 Local Related Literature

2.1.1 Filipinos Spent 1.54 Billion Seconds Using Digital Lending Apps in 2024

Studies on digital lending practices in the Philippines reveal a significant growth in the use of online loan applications and mobile lending platforms, indicating how digital channels are reshaping credit access and consumer behavior. Research and industry reports show that the adoption of digital lending apps in the Philippines increased sharply in recent years, with Filipinos spending substantial cumulative time engaging with these platforms — an indicator of growing reliance on digital credit mechanisms for personal loans and other financial services (Digido analysis, 2025; digital lending usage reports, 2024). These studies highlight how digital lending systems provide easier application processes and broader access compared to

traditional credit sources, suggesting that the convenience and accessibility of digital platforms are key drivers of increased usage. Moreover, analyses of user engagement with lending apps show a rise in unique users and active sessions, reinforcing the trend of Filipinos turning to digital loan applications as an alternative to traditional banking products (Philstar.com reports on lending app usage, 2024).

Despite these insights into adoption and usage trends, current local research often emphasizes consumer behavior and market growth rather than the technical aspects of digital loan application systems with structured data encoding or systematic payment monitoring. While reports document the surge in time spent on digital lending apps and growing user bases, they do not evaluate how these platforms manage loan records, repayment tracking, or organized database systems related to loan processing. Additionally, although these studies demonstrate increased usage, there is limited discussion on the effectiveness of digital monitoring features, user data security, and how these systems support accurate and organized record management within institutional settings. Consequently, there is a research gap in the Philippines literature focusing on the design, implementation, and evaluation of integrated digital loan application modules that incorporate direct data encoding and payment monitoring for reliable recordkeeping. Overall, while Philippine studies highlight expanding digital lending usage, future research must investigate the operational effectiveness and system design elements that support organized loan record management.

2.1.2 Digital Lending Usage in Philippines Surges 37%

Studies analyzing the digital lending landscape in the Philippines highlight how the rapid growth of mobile and online lending services reflects broader shifts in credit accessibility and financial behavior among Filipino consumers. Reports show that the use of digital lending applications in the Philippines surged significantly in recent years, with total usage time and app engagement rising markedly compared to previous years, indicating increasing consumer reliance on digital credit solutions especially for personal and emergency financial needs (Crowdfund Insider, 2024; Philippine Star, 2024). Research on leading digital micro-lending services further illustrates how platforms like GLoan—part of the GCash ecosystem—have become dominant in the local market by empowering users to access loans, manage credit histories, and improve financial literacy, particularly among underserved populations historically excluded by traditional banking channels (Daily Guardian, 2024). These studies collectively emphasize the role digital lending platforms play in enhancing convenience and expanding financial access across different segments of Filipino society.

Despite these valuable observations of usage trends and market dynamics, existing local studies often focus on high-level adoption statistics or market leadership rather than system design, loan application encoding, or structured payment monitoring functionality. While usage figures demonstrate the popularity of digital lending platforms, there are few empirical studies that investigate how these systems organize loan records, track repayment behavior, or support database-based management of loan applications and payments within financial institutions.

Additionally, research tends to center on consumer behavior and market penetration rather than the technical performance, data accuracy, or operational processes behind digital lending workflows. Consequently, there is a research gap in the Philippine literature concerning comprehensive evaluations of integrated digital loan application and payment monitoring modules with direct data encoding and record-keeping capabilities.

2.1.3 Grocery and Rice Trading Inventory Module

Inventory management challenges and the shift toward digital systems are recurring themes in retail research conducted in the Philippines. Several studies indicate that manual recording methods in small retail and food trading businesses lead to inaccurate stock counts, inefficient monitoring of daily sales, and operational bottlenecks (Baylen, 2020; Gabriel, 2025; Nadores et al., 2026). For example, research on inventory practices among small-sized restaurants in Quezon Province found that traditional stock monitoring contributed to discrepancies between actual and recorded inventory and reduced operational efficiency, suggesting the need for computerized systems (Baylen, 2020). Similarly, studies assessing grocery and rice store operations reported inconsistencies in purchasing, storing, and recording inventories, which hinder effective control of stock and sales transactions (Gabriel, 2025; Nadores et al., 2026). These findings collectively demonstrate a growing recognition of digital inventory technologies as tools that can streamline stock

monitoring, support real-time tracking of product movement, and generate accurate transactional reports in Philippine retail environments.

Despite these insights into inventory practices and the potential benefits of computerized systems, several limitations persist in the current literature. Most of the existing studies focus on manual inventory challenges and general recommendations for digitization, but they do not provide detailed evaluations of actual web-based inventory systems tailored to the needs of small retail or cooperative enterprises (Gabriel, 2025; Nadores et al., 2026). Additionally, there is limited empirical research on modular inventory platforms that integrate real-time sales recording with stock tracking across multiple product categories such as grocery items and rice products. For instance, while Baylen's (2020) work identifies inventory issues in restaurants, it does not examine how digital systems perform in collaborative trading environments or how such systems affect decision-making, user interaction, and operational efficiency in real-world settings. Consequently, a research gap exists in developing and assessing specialized inventory management modules designed for cooperative or multi-category retail contexts that can ensure accurate stock monitoring, efficient daily sales recording, and reliable reporting. Addressing this gap will support the design of web-based inventory solutions tailored for cooperative structures like the Bislig Chambers Multi-Purpose Cooperative, contributing to more effective and data-driven retail operations.

2.1.4 Digital Cooperative Management Systems

Studies on cooperative digitalization in the Philippines reveal a clear trend toward adopting information and communication technologies (ICT) to improve administrative efficiency and service delivery. Several reports indicate that Philippine cooperatives are increasingly embracing digital tools — such as web-based portals, POS systems, and online platforms — to enhance operational transparency, member access, and financial services (Asian Farmers Association for Sustainable Rural Development [AFA], 2024; Globe Business, 2024). For example, agricultural cooperatives in Region 13 installed modern POS and lending software systems to streamline sales tracking, bookkeeping, and reporting, which significantly reduced manual workload and improved operational accuracy (AFA, 2024). Similarly, Globe Business initiatives in partnership with the Cooperative Development Authority (CDA) have equipped cooperatives with digital transformation tools — including cloud services, collaboration platforms, and communication tools — enabling even rural cooperatives to adopt ICT and engage members more efficiently (Globe Business, 2024). These developments demonstrate a growing recognition that digital platforms can automate record-keeping, streamline financial transactions, and expand access to cooperative services, aligning with broader trends in Philippine financial digital transformation.

Despite the positive movement toward digitalization, current Philippine literature on cooperatives' digital transformation also highlights several limitations. Many reports and initiatives focus on technology adoption and capability building in general terms rather than conducting in-depth empirical evaluations of how specific

web-based cooperative management systems perform in practice. For instance, while POS and lending software deployments in Region 13 demonstrate practical adoption steps, they do not assess system usability, member engagement outcomes, or long-term operational impact (AFA, 2024). Similarly, discussions of ICT tools provided by Globe Business emphasize infrastructure and training support but lack detailed analysis of how these technologies integrate with cooperative workflows such as loan application processing, payment monitoring, and member database management (Globe Business, 2024). Moreover, research on digital transformation in multipurpose cooperatives identifies functions and challenges of IT integration, but it does not fully explore how a comprehensive web-based portal could centralize these functions into a unified management system (Rabanes & Janamjam, 2025). Consequently, there remains a research gap in examining holistic web-based cooperative information management systems that combine member services, loan tracking, inventory integration, and secure digital transaction processing — an area directly relevant to the proposed study on the Bislig Chambers Multi-Purpose Cooperative.

2.2 Local Related Studies

2.2.1 The Impact of Digitalization on Credit Risk Management in Microfinance Institutions in Nueva Ecija, Philippines

Studies conducted within the Philippines show growing interest in the digitalization of lending practices and credit risk management, highlighting how digital tools can enhance efficiency and operational performance in financial institutions. For

example, Campos (2024) found that digitalization in credit risk management practices among microfinance institutions in Nueva Ecija improves operational efficiency and offers opportunities to streamline loan collection policies and repayment tracking, even though traditional performance measures remain a challenge (Campos, 2024). Research into credit management systems developed for Philippine credit cooperatives indicates that digital platforms for managing loans, payments, member records, and reports significantly reduce data redundancy and improve record accuracy compared to manual processes, leading respondents to support their adoption for better transaction management (Cofino, 2021). Additionally, industry reports on the Philippines' expanding digital lending market reveal that user engagement with digital lending applications continues to rise, suggesting that Filipinos increasingly rely on digital loan platforms for personal credit access, further emphasizing the relevance of digital solutions in lending and financial recordkeeping.

Despite these insights, most Philippine studies do not fully explore integrated digital loan systems that combine direct loan application encoding, real-time payment monitoring, and structured database management. Findings from Campos (2024) highlight the potential of digitalization to improve efficiency, but there remains limited focus on detailed system design, usability, and long-term performance of integrated loan modules in microfinance and cooperative settings. Similarly, the credit management system research (Cofino, 2021) supports digital transaction handling but predates recent advances in fintech and does not address user adoption or scalability challenges in larger institutions. Reports on digital lending usage in the Philippines primarily describe market trends and user behavior rather than technical

system evaluations or monitoring outcomes. Consequently, there is a research gap in comprehensive local studies that evaluate holistic digital loan application and payment monitoring solutions tailored for Filipino institutions. Overall, while Philippine literature highlights the growing role of digital tools in lending and credit management, future research should focus on empirical evaluations of integrated digital modules with direct data encoding and systematic payment tracking to support effective loan oversight.

2.2.2 Inventory Management Practices of Retailers in Santiago City

The transition from manual to web-based inventory and sales systems has been widely recognized as a solution for improving accuracy in stock monitoring and sales recording in retail and trading environments. Several studies report that systematic inventory management enhances record accuracy, reduces discrepancies, and supports organized sales documentation. For instance, the study "Inventory Management Practices of Retailers in Santiago City" examined inventory practices among 50 small retailers including grocery and rice stores, and found that while purchasing practices were frequently implemented, record-keeping, storage, and issuance processes were seldom practiced, leading to inconsistent monitoring of stock and transaction data. The research highlights that effective inventory control requires accurate documentation and active monitoring, suggesting that structured systems could help bridge the gap between physical stock and recorded data. These findings underscore the need for web-based solutions to ensure precise

inventory-tracking and efficient daily sales recording—core components necessary for the grocery and rice trading module of this study

Despite demonstrated improvements, existing studies on inventory management practices often reveal persistent challenges in accurate record-keeping and structured daily transaction documentation, particularly among small retailers that rely on manual or informal processes. The "Inventory Management Practices of Retailers in Santiago City" research highlights that although retailers frequently engaged in purchasing activities, they rarely updated records daily or maintained systematic storage and issuance logs, which contributed to moderate problems in inventory accuracy and operational control. This suggests a gap in systems that integrate comprehensive inventory tracking with structured sales recording, especially for multi-category products like groceries and rice that require regular updates and precise data handling. Addressing this gap supports the study's objective to develop a web-based grocery and rice trading module that ensures reliable stock monitoring, precise transaction recording, and improved operational efficiency for the Bislig Chambers Multi-Purpose Cooperative.

2.2.3 Data Driven Point of Sale and Inventory System for Pastil Sa Tabi: Integrating Sales Forecasting Algorithms and Predictive Analytics

The transition from manual to web-based inventory and sales systems has been widely recognized as an effective solution for improving accuracy in stock monitoring and sales recording in retail and trading environments. Several recent

studies highlight the importance of integrating inventory tracking with point-of-sale functions and predictive analytics to enhance operational efficiency and data reliability. For example, the study “Data-Driven Point-of-Sale and Inventory System for Pastil Sa Tabi: Integrating Sales Forecasting Algorithms and Predictive Analytics” described a web-based application that combines sales transaction processing, inventory management, and demand forecasting to automate previously manual processes, resulting in real-time analytics, increased accuracy in inventory updates, and improved decision-making. The system was evaluated positively for its ability to record transactions and generate reliable sales and inventory reports, demonstrating how web-based solutions can improve data accuracy and operational speed in small business contexts. These findings underscore the relevance of data-driven and automated systems in ensuring precise inventory tracking and efficient daily sales recording—key elements needed for the grocery and rice trading module of this study.

Despite the demonstrated benefits of integrating predictive analytics with sales and inventory management, existing research often focuses on performance improvements and forecasting capabilities without fully addressing the varied operational needs of multi-category product trading environments such as groceries and rice. While the Pastil Sa Tabi system improved accuracy and automation in a food enterprise context, it did not explore how such systems can be adapted to handle the specific complexities of diverse product types, units of measure, and fluctuating pricing structures commonly found in cooperative trading operations. Moreover, studies on data-driven point-of-sale systems tend to emphasize technical functionality and user evaluations rather than integration with broader organizational workflows or

daily sales record structuring for cooperative stakeholders. This highlights a gap in the literature for systems capable of combining precise inventory control with comprehensive daily sales documentation in multi-category trading contexts. Addressing this gap supports the study's objective to develop a web-based grocery and rice trading module that ensures reliable stock monitoring, accurate transaction recording, and enhanced operational efficiency for the Bislig Chambers Multi-Purpose Cooperative.

2.2.4 Development of Financial Management System for a Barangay Cooperative

Cooperative organizations traditionally rely on manual and paper-based processes for tracking membership details, financial transactions, loan management, and other administrative tasks, which can be time-consuming, error-prone, and inefficient. To address these challenges, several studies have explored the design and implementation of web-based systems tailored for cooperative environments. For instance, a study by Aligno et al. (2025) developed a Web-Based Financial Management System for the Mantibugao Agrarian Reform Beneficiary Farmers Cooperative (MARBFC), which replaced manual financial record-keeping with an automated online solution that streamlined membership processing, loan tracking, share capital management, and revenue reporting. The researchers employed the Waterfall methodology for system development, conducted usability testing using the System Usability Scale (SUS), and reported enhancements in operational

transparency, organizational efficiency, and decision-making support for cooperative staff and members.

This research underscores the potential of web-based systems to improve cooperative information management by automating routine processes, reducing errors, and facilitating faster access to accurate data. The MARBFC system's positive outcomes—such as improved procedural speed and better organized financial records—suggest that similar web-based information management solutions can be effectively adapted to other cooperative contexts like BICHAMCO. Developing a tailored web-based cooperative loan and information management system for Bislig Chambers Multi-Purpose Cooperative would continue this trend of digital transformation in cooperatives, by integrating membership administration, transaction processing, reporting, and other core functions into a unified platform. These solutions thus demonstrate how information technology can support cooperative sustainability and member service delivery in diverse local settings.

2.2.5 Credit Management System: An Effective Tool for Credit Cooperatives in the Philippines

A study titled "Credit Management System: An Effective Tool for Credit Cooperatives in the Philippines" by Chester L. Cofino (2021) examined the design and implementation of a credit management system intended to serve as an automated platform for credit cooperative operations. The research employed the Waterfall Model as part of the software development life cycle (SDLC) to guide the

systematic design, development, and evaluation of the system. The resulting application was capable of managing key tasks such as member record keeping, processing loan applications, generating amortization schedules, tracking loan payments, and producing reports on collections and disbursements. To assess its effectiveness, the study used survey questionnaires among cooperative staff and members and found that respondents favored system adoption as a means to achieve better-managed transactions and more organized credit records.

The findings from Cofino's study highlight the practical benefits of introducing computerized credit management systems in cooperative settings, including improved processing speed, reduced data inconsistencies, and enhanced ability to monitor credit portfolios. These outcomes support the broader trend of digital transformation in cooperative management by demonstrating how information systems can replace manual procedures and contribute to operational effectiveness. In the context of developing a web-based cooperative loan and information management system for Bislig Chambers Multi-Purpose Cooperative (BICHAMCO), integrating a credit management module can build on these insights by offering real-time processing, centralized data access, and automated reporting tools that align with the cooperative's administrative and service delivery goals.

2.3 Foreign Related Literature

2.3.1 Digital Credit and Loan Management Systems

The increasing adoption of digital financial technologies in the Philippines has reshaped how loans are applied for, processed, and monitored, aligning with broader global trends in fintech innovation. Several Philippine based reports and studies point to rapid growth in digital lending platforms, highlighting how these systems enable remote loan application processing, automated approvals, and real time repayment tracking, thereby improving efficiency and accessibility in credit management (Bangko Sentral ng Pilipinas [BSP], 2023; Philippine FinTech Report, 2024). For instance, the BSP's national survey of digital lenders documented that Philippine digital credit operators use automated loan decisioning, dynamic credit scoring, and machine learning models to reduce processing times and mitigate manual errors in credit evaluations (BSP, 2023). Similarly, analyses of digital lending trends in the country show that real time monitoring tools enhance transparency and allow lenders to better organize borrower information and repayment histories (Philippine FinTech Report, 2024). These findings demonstrate that web based and mobile enabled loan platforms are increasingly recognized for their potential to streamline credit operations and support structured financial recordkeeping in the Philippine financial ecosystem.

Despite these developments, existing Philippine literature and industry reports largely focus on macro level fintech adoption, regulatory frameworks, and consumer behavior rather than the design and performance of custom, institution specific digital loan management systems. Many investigations emphasize

regulatory compliance, fraud prevention, and digital credit expansion (BSP, 2023; Philippine FinTech Report, 2024), but they do not fully examine how integrated digital platforms can be adapted for cooperatives and locally governed financial institutions with unique transaction workflows and member based structures. Furthermore, there is limited empirical evidence on how digital systems can be tailored to support direct data encoding, systematic loan monitoring, and organized repayment tracking in cooperative settings. This reveals a gap in the literature concerning the operational effectiveness of digital credit modules designed for institutions like Bislig Chambers Multi Purpose Cooperative, which need solutions that integrate loan application processing, record management, and real time repayment monitoring. Addressing this gap supports the proposed study's objective to develop a web based digital loan application and payment monitoring system that meets the specific cooperative operational requirements.

2.3.2 Web-Based Inventory Systems for SMEs in E-Commerce

The transition from manual to web-based inventory and sales systems has been widely recognized as an effective solution for improving accuracy in stock monitoring and sales recording in retail and trading environments. Several studies report that web-based inventory information systems significantly streamline inventory control, reduce errors inherent in manual processes, and support organized sales documentation. For example, the research "Web-Based Inventory Systems for SMEs in E-Commerce" highlighted the increasing relevance of online inventory

management systems for small and medium enterprises (SMEs) operating in digital marketplaces, demonstrating that web-based platforms improve real-time stock visibility and offer centralized control over inventory data while supporting efficient data access and reporting. Other research similarly finds that web-based systems enable SMEs to better manage stock across multiple channels, reduce discrepancies between physical and recorded inventory, and enhance operational workflows by integrating sales and inventory data on a unified platform. These findings highlight the potential of web-based solutions to ensure precise inventory-tracking and efficient daily sales recording—core components necessary for the grocery and rice trading module of this study.

Despite demonstrated improvements, existing web-based inventory information systems often emphasize general inventory control and broad e-commerce needs without addressing challenges specific to multi-category goods trading, such as rice products that may require different units of measure, dynamic pricing models, and frequent stock turnover. The “Web-Based Inventory Systems for SMEs in E-Commerce” research points to the benefits of web-based platforms for online SMEs but focuses primarily on the general advantages of real-time tracking and enhanced data access, with limited examination of how such systems can be tailored to support specialized trading modules combining diverse product types and detailed transaction logs. Moreover, the study does not deeply explore how web-based systems integrate inventory tracking with daily sales recording in cooperative or multi-purpose organizational contexts, revealing a gap in the literature for systems designed to handle both intricate stock monitoring and structured sales

documentation within varied trading operations. Addressing this gap supports the study's objective to develop a web-based grocery and rice trading module that ensures reliable stock monitoring, precise transaction recording, and improved operational efficiency for the Bislig Chambers Multi-Purpose Cooperative.

2.3.3. Analyzing Grocery Items Inventory and Managing Stock

Recent studies on inventory management indicate that the adoption of computerized systems significantly enhances operational accuracy, stock monitoring, and sales documentation in retail environments. For example, Baggyalakshmi, Anushree, and Revathi (2023) found that integrating Point of Sale (POS) systems with real-time inventory tracking allows businesses to prevent stockouts and overstocking by automatically recording sales transactions and monitoring product movements. Their research highlights key features such as automatic sales recording, expiration date tracking, and detailed inventory reporting, which together support more informed operational decision-making and reduce the likelihood of human error. These findings align with broader trends in retail technology adoption in the Philippines, where small and medium enterprises are increasingly investing in digital inventory tools to streamline daily operations, improve data reliability, and facilitate faster access to inventory and sales information. Such technologies are therefore positioned as essential for improving the efficiency and transparency of stock management processes in contemporary grocery and retail settings.

Despite the demonstrated benefits of computerized inventory systems, existing research also reveals limitations that are particularly relevant for cooperative trading environments. The study by Baggyalakshmi et al. (2023), while informative about general grocery inventory practices, focuses primarily on commercial retail contexts and does not examine unique product categories such as rice, which involve distinct storage requirements, pricing structures, and demand patterns. Furthermore, the research does not consider how inventory systems can be customized for cooperative settings, where operational workflows, multi-user coordination, and shared ownership structures differ from those in private supermarkets. This suggests a gap in the literature regarding inventory modules that support not only general grocery inventory processes but also the specific needs of cooperatives handling both grocery and rice products. Addressing this gap is crucial for developing a web-based inventory management system capable of accurate stock tracking, daily sales recording, and integrated reporting tailored to institutions like the Bislig Chambers Multi-Purpose Cooperative.

2.3.4 Mobile-Friendly Lending Portal Driving Higher Borrower Satisfaction and Employee Efficiency

The evolution of digital lending technologies has increasingly emphasized user-centric, mobile-friendly interfaces that improve borrower accessibility and streamline credit processes across devices. Several industry analyses and case reports highlight that responsive web portals and mobile lending platforms enable

users to apply for loans, monitor application status, and make secure repayments without the friction associated with traditional branch-based services (ScienceSoft, n.d.; FinTech Global, 2024). For instance, the ScienceSoft case study on a mobile-friendly lending portal demonstrated the integration of features such as real-time loan status tracking, account management tools, secure online payments, and notification systems, all accessible through both desktop and mobile browsers. According to the report, these responsive features enhanced overall borrower satisfaction and improved operational efficiency by reducing manual workloads for staff and providing a convenient digital experience for users. Such trends reflect a broader shift within financial services toward responsive, device-agnostic solutions that support remote access and real-time interaction with credit products and services.

Despite these positive themes, existing literature on mobile-friendly lending portals often focuses on general financial service providers and commercial banks or fintech platforms, with limited attention to how such systems function in community-based or member-centric organizations like cooperatives. The ScienceSoft case study, for example, effectively highlights enhanced usability and integration benefits but does not explore cooperative-specific functionalities such as member governance integration, role-based access, or how digital portals can support loan application, monitoring, and payment workflows within the cooperative structure (ScienceSoft, n.d.). Moreover, the case lacks empirical evidence on long-term user adoption, satisfaction metrics, and comparative performance against traditional cooperative processes. This indicates a gap in the literature regarding comprehensive evaluations of mobile-friendly lending systems that are tailored for community

institutions where social capital, collective decision making, and varied member roles influence system usage. Addressing this gap would contribute to a better understanding of how digital loan portals can be adapted to support the unique needs of cooperative members and stakeholders, aligning with the objectives of this study on the Bislig Chambers Multi Purpose Cooperative.

2.4 Foreign Related Studies

2.4.1 Digital Transformation in Financial Lending Platforms

Studies on digital financial platforms increasingly highlight how digital loan origination and management frameworks improve accessibility, reduce manual errors, and enable real-time tracking of loan applications and repayments (Liu et al., 2025; international research on digital lending platforms, 2024). Research focused on fintech transformation demonstrates that digital platforms integrating automation, risk assessment, and payment tracking contribute to faster decision-making, improved customer experience, and operational efficiency in lending services. For example, literature on smart financial management systems underscores the significance of web-based interfaces and integrated payment modules in managing loan disbursement, installment tracking, and secure data storage in financial cooperatives and institutions (Hakiki et al., 2025). Additionally, systematic reviews in the fintech domain affirm broader adoption of digital channels for lending and financial services, aligning digital innovation with efficiency and user convenience.

Many studies on digital lending systems tend to focus broadly on fintech adoption and technological innovation without rigorously testing end-to-end loan application and payment monitoring modules in real institutional settings. While systematic reviews and fintech analyses identify trends in digital transformation, they often lack detailed insights into how specific features—such as direct encoding interfaces and structured payment tracking—perform in operational contexts. Furthermore, although research on smart financial management systems (e.g., automated web applications) demonstrates functionality and integration benefits, these studies primarily assess technical performance rather than user adoption challenges, scalability issues, or security risks inherent in digital loan ecosystems. There remains limited empirical evaluation of digital loan modules specifically designed to handle high-volume transactional data and long-term record management across diverse organizational environments.

Consequently, there is a research gap in comprehensive, empirical studies that evaluate holistic digital loan application systems with direct data encoding, automated payment monitoring, and structured record management—not just as proof-of-concepts but through real deployment scenarios. Future research should address how such systems can balance performance, security, usability, and sustainability in dynamic lending environments, particularly in developing economies with limited digital infrastructure. Overall, while current literature supports the digital transformation of loan processing and monitoring, future research must deepen empirical evidence on integrated system design and real-world operational effectiveness to fully validate these innovations' impact.

2.4.2 Smart Financial Management for Cooperatives

Several studies indicate that implementing web-based and automated loan management systems significantly improves data accuracy, transparency, and transaction monitoring (Miguel et al., 2025; Palang et al., 2025). Similarly, research on integrated financial platforms demonstrates that linking digital loan applications with real-time payment tracking strengthens reporting capabilities and reduces administrative workload (Hakiki et al., 2025). Furthermore, studies on loan origination and digital record processes emphasize that structured database systems and automated documentation enhance operational performance and support better decision-making (Nasution et al., 2024). Complementing these findings, investigations into digital lending applications stress the importance of secure data handling and systematic monitoring to ensure compliance and maintain borrower trust (Akanji et al., 2026).

However, despite the documented advantages of digital loan application and payment monitoring systems, several limitations remain evident in the existing literature. While studies by Miguel et al. (2025), Palang et al. (2025), and Hakiki et al. (2025) demonstrate improvements in efficiency and record organization, many of these investigations are limited to specific institutions such as cooperatives or village enterprises, reducing their generalizability to other organizational settings. Additionally, some studies focus primarily on system functionality and user satisfaction metrics without deeply examining long-term system sustainability, data security vulnerabilities, or scalability concerns. Research by Nasution et al. (2024) emphasizes digital documentation processes but does not fully explore user adoption

challenges in smaller institutions with limited technical resources. Moreover, although Akanji et al. (2026) highlight privacy and compliance issues, there remains limited discussion on how structured monitoring modules can be customized for localized contexts, particularly in developing countries.

The existing literature therefore lacks comprehensive studies that integrate direct encoding, systematic payment monitoring, secure record management, and contextual adaptability within a single modular framework. Specifically, there is a research gap in designing and evaluating a fully integrated digital loan application and payment monitoring module tailored for institutions requiring organized record management with practical implementation considerations. Overall, while current research supports the effectiveness of digital loan systems, further studies are needed to develop scalable, secure, and context-sensitive solutions that address operational and technological constraints in diverse environments.

2.4.3 Web-Based Inventory and Sales Information Systems for MSMEs

Research on web-based information systems highlights the importance of digital platforms in improving operational efficiency, data management, and service delivery in businesses, particularly for Micro, Small, and Medium Enterprises (MSMEs). Studies indicate that implementing web-based inventory and sales systems enables organizations to integrate data, automate processes, and provide real-time access to financial and operational information (Angellin et al., 2023). For example, the case study on Cik Cik Meatshop and Frozen Foods in Indonesia

demonstrated that a web-based system significantly improved inventory management, minimized errors, and streamlined sales reporting. These findings suggest that digital systems not only enhance organizational efficiency but also empower users to access and monitor business transactions accurately and in real time, which aligns with the broader trend of technology-driven solutions for service delivery and management.

While the adoption of web-based systems provides substantial benefits, certain challenges remain in ensuring full utilization and effectiveness. The study identified that businesses initially relying on manual processes faced issues such as errors in transaction records, inefficiencies in inventory control, and delays in reporting (Angellin et al., 2023). These limitations underscore the need for accessible, user-friendly online platforms that allow stakeholders to manage data efficiently and perform tasks such as transaction monitoring, report generation, and real-time updates. Developing a member or user portal capable of handling online transactions, monitoring records, and managing inventory or financial information can address these gaps by improving accessibility, enhancing operational efficiency, and providing users with immediate and accurate information. Such systems are critical for modern business operations and can serve as a model for digital transformation in cooperative and financial institutions.

2.4.4 The Role of Cooperative Banks in Promoting Financial Inclusion and Access to Credit in Underserved Communities in Enugu State

Research on cooperative banking and financial inclusion highlights the growing role of technology in improving access to financial services for underserved communities. Studies indicate that cooperative financial institutions help expand access to credit and financial resources, especially in areas where traditional banking services are limited. For instance, the study conducted by Ugwuanyi Lilian and Osita Bertrand Obodoechi (2024) found that many cooperative members experienced improved access to financial services after joining cooperative banks. Their findings also emphasized the importance of technological solutions that can enhance service delivery and improve accessibility to financial services. These results suggest that digital platforms and online systems can strengthen cooperative operations by allowing members to conveniently access services such as loan applications, payment processing, and transaction monitoring.

However, despite the benefits of cooperative banking in promoting financial inclusion, several operational challenges still limit the efficiency of service delivery. The study by Ugwuanyi Lilian and Osita Bertrand Obodoechi (2024) revealed that some members face barriers such as limited financial resources, lack of awareness of available services, and insufficient technological infrastructure. These challenges indicate that many cooperatives still rely on traditional or manual processes that may slow down financial transactions and limit accessibility. Consequently, the lack of integrated digital systems creates a gap in providing convenient financial services to members. Developing a member portal that allows members to apply for loans,

monitor transactions, and make payments online can help address this gap by improving accessibility, transparency, and operational efficiency. Such technological solutions can further strengthen the role of cooperatives in delivering accessible and efficient financial services.

2.4.5 Digital Cooperative Strategies to Enhance Member Services

Research on digital cooperatives highlights the increasing importance of technological integration in improving cooperative services and enhancing benefits for members. Studies suggest that digital applications enable cooperatives to streamline operations, increase transparency, and facilitate greater member participation in financial activities. For instance, research on digital cooperatives indicated that the use of digital systems in daily operations allows organizations to manage services more efficiently and provide substantial benefits to their members (Ermaya et al., 2021). These findings demonstrate that integrating digital tools into cooperative management contributes to improving members' economic welfare while reinforcing organizational efficiency. Such technology-driven systems are essential for enhancing the delivery of financial and administrative services within cooperatives.

While digital transformation offers significant advantages, several operational challenges continue to limit its full effectiveness. The study identified obstacles such as limited member capital, insufficient technological knowledge among personnel, weak technical skills, and low interest among younger generations in cooperative businesses (Ermaya et al., 2021). These constraints indicate that many cooperatives

have not yet fully optimized the potential of digital technologies. Consequently, the absence of accessible digital platforms creates a gap in delivering efficient cooperative services. Implementing a member portal that allows members to apply for loans, monitor transactions, and make online payments can address this gap by enhancing accessibility, improving operational efficiency, and promoting greater member engagement. Such digital solutions are pivotal for modernizing cooperative systems and strengthening their capacity to provide effective financial services.

Chapter III

TECHNICAL BACKGROUND

This chapter discussed the technical specification and design of the proposed application system needed for development

3.1 Technical Study and Feasibility

The development of a web-based Cooperative Loan and Information Management System for Bislig Chambers Multi-Purpose Cooperative (BICHAMCO) is necessary to address the inefficiencies caused by manual and fragmented processes in managing loans, inventory, member transactions, and reporting. Manual record-keeping often results in delays, data inconsistencies, and human errors that affect the accuracy and reliability of cooperative operations. To improve efficiency and organization, the proposed system will construct a digital loan application and payment monitoring module that enables direct encoding and organized record management, ensuring accurate and accessible loan information. It will also develop a web-based grocery and rice trading module to support accurate inventory tracking and daily sales recording, allowing staff to efficiently monitor stock levels and transactions. Additionally, the system will establish a Member Portal that enables members to apply for loans, view loan details, monitor transactions, and make payments online through GCash, improving accessibility and convenience. Furthermore, the system will implement an integrated reporting feature that generates accurate and timely reports to support better decision-making and

effective operational monitoring. Overall, the proposed system will provide BICHAMCO with a centralized, reliable, and efficient platform that enhances productivity, minimizes human error, and improves the overall management of cooperative operations.

3.1.1 Hardware Requirement and Specification

During the implementation of the proposed system, the following hardware requirements and specifications will be observed. The minimum hardware requirements for the proposed system include, but are not limited to, its development and deployment environments.

Table 1. Hardware Requirements and Specification

Description	Development	Deployment
Processor	Intel Core i5 10th Generation or higher	Intel Core i5 or higher
Memory	RAM 8GB	RAM 4GB
Storage	HDD/SSD 256GB	HDD/SSD 128GB
Operating System	Windows 10/11	Windows 10/11
Internet Access	Required	Required
Peripherals	Keyboard, Mouse	Keyboard, Mouse

Table 1 presents the hardware requirements for both development and end-user usage of the proposed web-based Cooperative Loan and Information Management System for BICHAMCO. Since the system is web-based, both development and deployment require a stable internet connection to ensure reliable access, data synchronization, and system availability. The development computer must be capable of handling multiple tasks simultaneously, including coding, testing, database management, and system deployment. Likewise, end-user devices should have sufficient processing capability and internet connectivity to allow staff and members to efficiently access system features such as loan management, inventory tracking, transaction monitoring, and report generation.

3.1.2 Software Requirement and Specification

During the development of the Cooperative Loan and Information Management System for BICHAMCO, the proponents utilized specific software requirements and their corresponding specifications to ensure efficient system design, development, testing, and deployment. These software tools supported essential development processes such as coding, database management, interface design, and system testing. The selected software was reliable, user-friendly, and compatible with web-based technologies, ensuring that the system can function effectively and be maintained with ease.

Table 2. Software Requirements and Specification

Component	Software Used
Front-End	HTML, CSS, JavaScript
Back-End	PHP, PHPMailer
Database Management	MySQL
Server Environment	XAMPP (for local deployment) Hostinger (for online deployment)
Web Browser	Google Chrome / Microsoft Edge
Document Handling	Excel / CSV File Import/Export

Table 2 presents the software requirements for the Cooperative Loan and Information Management System for BICHAMCO. The system utilizes HTML, CSS, and JavaScript for front-end development to create a responsive and user-friendly interface, while PHP is used for server-side processing to handle system logic and transactions. MySQL is implemented for database management to ensure secure and organized storage of member records, loan information, inventory data, and transaction history. XAMPP is used as the local development and testing environment, while Hostinger is utilized for online deployment to make the system accessible via the internet. In addition, web browsers such as Google Chrome and Microsoft Edge are supported for system access, and Excel/CSV file handling is incorporated to allow efficient data import and export for reporting and record management purposes.

3.1.3 User Requirement and Specification

The system requires only basic computer literacy from users, focusing on intuitive operations that minimize training time. Users must be able to navigate web browsers, understand form-based data entry, and perform basic file operations.

Basic User Requirements:

- Ability to use keyboard and mouse for data entry
- Understanding of basic web browser navigation
- Familiarity with Windows file system for document uploads/downloads
- Capability to read and interpret system-generated reports
- Basic understanding of financial processes

User Roles and Capabilities:

Administrator: The Administrator/Manager oversees the entire system and holds full access to all functionalities. This role is responsible for managing member registration, loan approvals, inventory management, staff accounts, system security, and data backup. Additionally, the Administrator/Manager generates consolidated reports for monitoring and auditing purposes and ensures the proper use and compliance of all system operations.

Loan Officer: The Loan Officer is tasked with managing all aspects of loan operations, including the registration of loan applications, approval of loans, recording of payment collections, maintaining

amortization schedules, and updating loan records. The Loan Officer ensures that all loan information is accurate and properly maintained.

Bookkeeper: The Bookkeeper handles the cooperative's financial records, monitoring all transactions related to loans as well as grocery and rice trading. This role prepares financial statements and reports, tracks balances, penalties, payments, and overall financial activity, and ensures the accuracy and integrity of the cooperative's financial data.

Cashier: The Cashier manages the inventory and daily sales transactions, recording member purchases whether in cash or on credit, updating inventory records, and assisting members with any transaction-related concerns.

Personal Collector: The Personal Collector is responsible solely for recording member loan payments, which are then forwarded to the Loan Officer for verification and pre-confirmation. Access for this role is limited strictly to payment recording.

Member: The Member can apply for loans and upload the required pre-approval documents. Members can view their own loan balances, due dates, penalties, and payment history, make loan payments online, and review their grocery and rice purchase history. Access for members is strictly limited to their personal records.

3.1.4 Network Requirement and Specification

Since the proposed system is a web-based application, an internet connection is required for the BICHAMCO Administrator, Officers, Front Desk personnel, Cashier, Bookkeeper, and Members to access the system. The BICHAMCO Loan and Information Management System is hosted on the Internet, allowing authorized users to securely access and manage member records, loan transactions, sales, and other cooperative services.

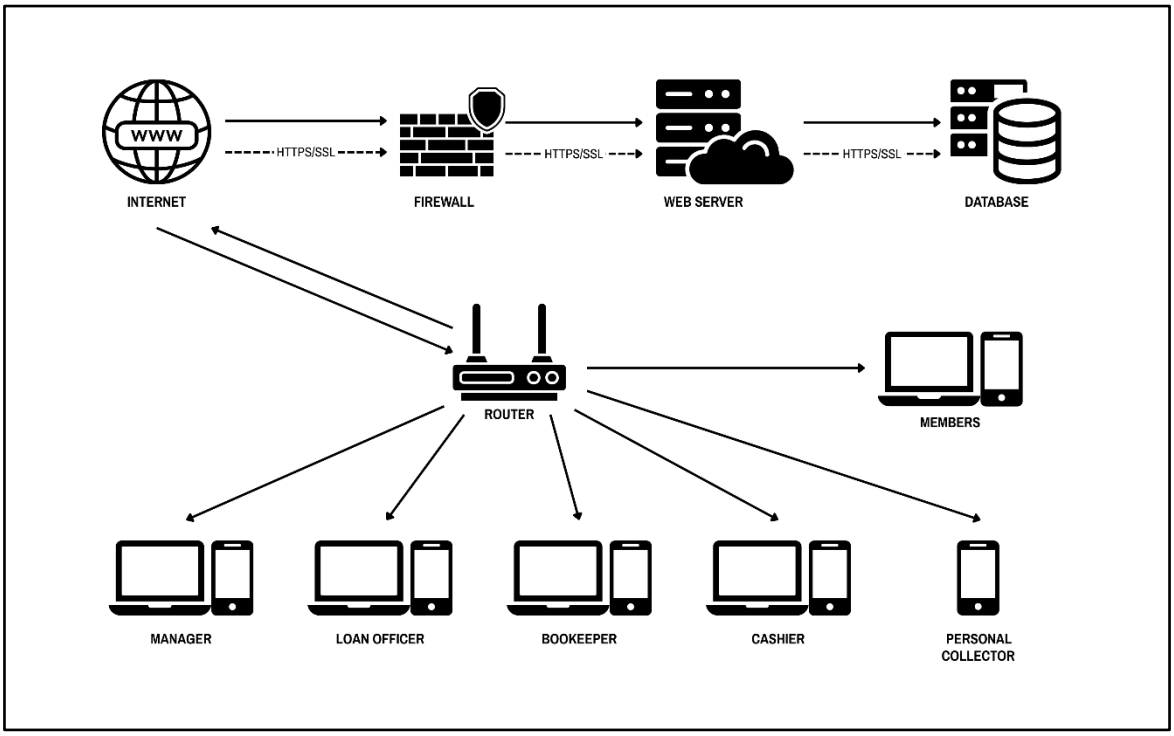

Figure 2. Network Infrastructure

The figure above outlined process of defining network requirements and specifications to create a reliable infrastructure. It discussed techniques for gathering, analyzing, and documenting

requirements and emphasized the importance of understanding the needs to meet the organizational goals.

3.1.5 System Architecture

The BICHAMCO Cooperative Loan and Information Management System follows a web-based client–server architecture that allows users to access the system through internet-connected devices such as computers or mobile browsers. In this architecture, users interact with the system through the web interface, while the application server processes user requests and handles system functionalities including loan management, inventory monitoring, member transactions, and report generation. The system communicates with a centralized database where all cooperative records and transaction data are securely stored. This architectural structure enables efficient data processing, centralized data management, and reliable system performance, ensuring that cooperative operations are organized and accessible to authorized users.

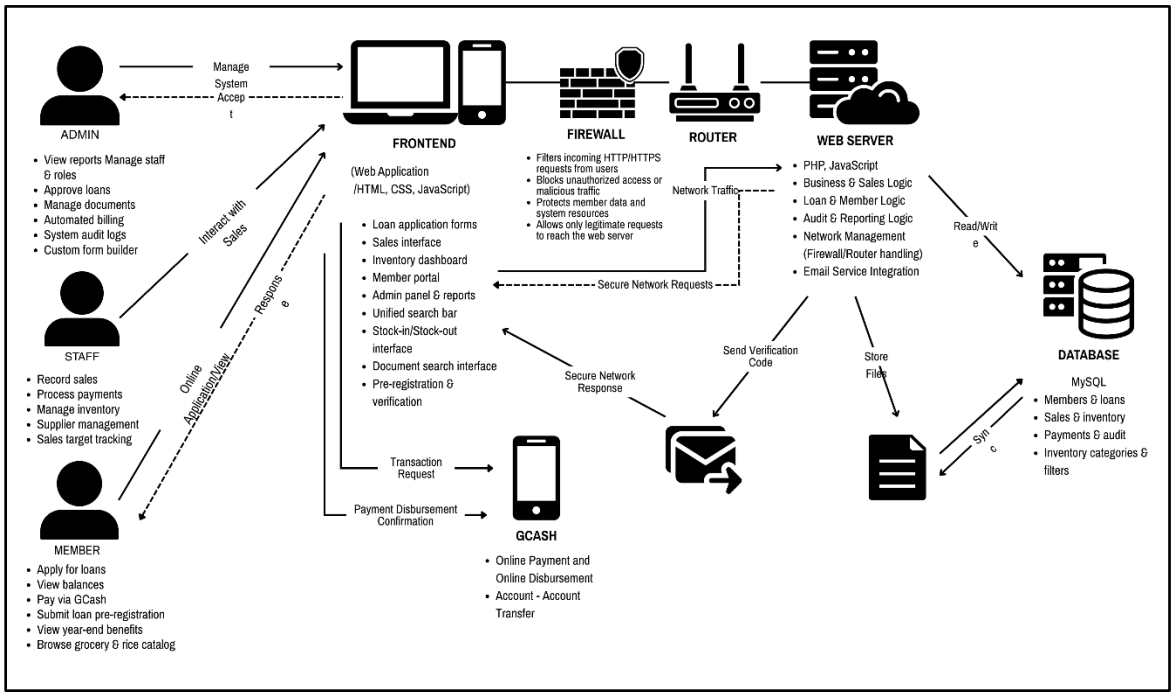

Figure 3. BICHAMCO Cooperative Loan and Information Management System Architecture

The proponents used login authentication as a security measure to control and verify access to the Cooperative Loan and Information Management System. Upon accessing the system, users are required to enter a valid username and password through the login page, which are then verified against stored credentials in the database. The authentication server processes the submitted information using PHP session logic and password hashing to confirm the user's identity. If the credentials match, the system grants access and directs the user to their role-based dashboard, whether as an Admin/Owner, Staff/Cashier, or Member. Otherwise, access is denied and the user is notified of the invalid entry. This mechanism ensures that only authorized individuals can access the system and its features according to their designated roles.

3.2 System Security

System security is an essential component of the proposed Cooperative Loan and Information Management System to ensure that all data and system operations are protected from unauthorized access, data breaches, and potential system failures. The system implements multiple security measures to safeguard sensitive cooperative information such as member records, loan details, and financial transactions. Database security mechanisms are applied to control user access and maintain the integrity of stored data, while regular weekly backups are conducted to prevent data loss and ensure system recovery in case of technical issues. The application system also requires secure user authentication through a login process with validated usernames and passwords, preventing unauthorized users from accessing the system. In addition, the system utilizes encryption technologies such as SHA-256 hashing for password protection and AES-256 encryption for sensitive database information. These security implementations ensure that all cooperative data remains confidential, secure, and accessible only to authorized users, thereby maintaining the reliability and trustworthiness of the system.

3.2.1 Database Security

Database security is a top priority, as it forms the foundation of the entire cooperative operation. The proponents ensured that users could only access information in a controlled manner without compromising data integrity. Security measures such as AES-256 encryption were implemented to protect sensitive information from internal and external attacks. Furthermore, weekly backups ensure recovery from recent changes to prevent data loss in case of system failures. All database users are properly verified and authorized before access.

3.2.2 Application System Security

The system employs a login operation that requires users to enter a valid username and password to access the application, preventing unauthorized access. The system is hosted on Hostinger, which provides built-in security features such as SSL encryption, firewall protection, and automatic daily backups. All data transmitted between clients and the server is protected via HTTPS using SSL/TLS encryption, ensuring data security during transmission. Additionally, passwords must meet the following complexity rules to strengthen security:

- At least 8 characters in length
- One uppercase letter (A–Z)
- One lowercase letter (a–z)
- One number (0–9)
- One special character (e.g., @, #, $, %, &)

This ensures that all accounts are protected against common password attacks.

3.2.3 Encryption

The BICHAMCO Cooperative Loan and Information Management System is a web-based application with role-based access. Only authorized administrators, officers, front desk personnel, cashiers, bookkeepers, and registered members can access the system according to their assigned roles. To protect login credentials and sensitive data, the system uses SHA-256 hashing with salt for passwords, making it practically impossible for attackers to reverse-engineer the information.

Additionally, AES-256 encryption is applied to sensitive database fields such as loan amounts, member personal information, and financial records, ensuring a high level of confidentiality and protection against unauthorized access.